

ODE

U1
§1 { 常微分方程: 单自变量
偏微分 ... : 2/2+ 自变量

微分方程中未知函数的最高阶数:
微分方程的阶

$$\text{一阶: } F(x, y, \frac{dy}{dx}) = 0 \\ \Rightarrow \frac{dy}{dx} = f(x, y)$$

$$(a, b) \quad F(x, \varphi(x), \varphi'(x)) \equiv 0$$

$y = \varphi(x)$ 为一个解.

(a, b) 定义区间

$\Phi(x, y) = 0$ 隐函数,

定理

$$\frac{dy}{dx} = f(x, y)$$

条件 $y|_{x=x_0} = y_0$ $\leftarrow$ 初值条件

$f(x, y)$ 在 $D: |x - x_0| \leq a, |y - y_0| \leq b$ 连续,

有连续偏导 $f_y'(x, y)$.

在 $|x - x_0| \leq h$

于唯一 $y = \varphi(x)$ 满足

特解

其中 $h = \min(a, m)$

$$M = \max_{(x,y) \in D} |f(x,y)|$$

§2

一. 可分离变量.

$\frac{dy}{dx} = \varphi(x)\phi(y)$ 可分离变量微分方程

$$\frac{dy}{\phi(y)} = \varphi(x) dx$$

积分

$$\frac{dy}{dx} = \frac{y(1-x)}{x}$$

$$\frac{dy}{y} = \frac{1-x}{x} dx$$

$$\ln|y| = \ln|x| - x + C_1$$

$$|y| = |x| e^{-x+C_1} \quad C = \pm e^{C_1}$$

$$y = Cx e^{-x}$$

二. 齐次

$$f(x,y) = g\left(\frac{y}{x}\right)$$

$$\frac{dy}{dx} = g\left(\frac{y}{x}\right)$$

零齐次微分方程.

$$u = \frac{y}{x} \quad y = ux$$

$$\frac{d(ux)}{dx} = g(u)$$

$$x \frac{du}{dx} + u = g(u)$$

$$\frac{du}{dx} = \frac{g(u) - u}{x}$$

$$\frac{du}{g(u) - u} = \frac{dx}{x}$$

$$\int \frac{du}{g(u) - u} = \ln|x|$$

eg. $\frac{dy}{dx} = \frac{xy}{x^2 + y^2}$

$$u = \frac{y}{x} \quad y = ux$$

$$x \frac{du}{dx} + u = \frac{u}{1+u^2}$$

$$x \frac{du}{dx} = \frac{-u^3}{1+u^2}$$

$$-\frac{1-u^2}{u^3} du = \frac{dx}{x}$$

变量代换法.

$$\frac{1}{2u^2} - \ln|u| = \ln|ax|$$

$$\frac{x^2}{2y^2} - \ln\left|\frac{y}{x}\right| = \ln|cx|$$

$$\frac{x^2}{2y^2} = \ln|cy|$$

$$y = ce^{\frac{x^2}{2y^2}}$$

§3 — 一阶线性微分方程

$$\frac{dy}{dx} + p(x)y = f(x)$$

s.t. $f(x) \equiv 0$.

$$\frac{dy}{dx} + p(x)y = 0$$

$$e^{-\int p(x)dx} \left(\int f(x) e^{\int p(x)dx} dx + C \right)$$

$$\frac{dy}{y} = -p(x)dx$$

$$p(x)u \square$$

$$p(x)y$$

$$\ln|y| = \int p(x)dx + \ln|c|$$

$$y = u e^{-\int p(x)dx}$$

令 $u = u(x)$ 为待定函数

$$\frac{dy}{dx} = e^{-\int p(x)dx} \frac{du}{dx} - u e^{-\int p(x)dx} p(x) \quad \text{代回}$$

$$e^{-\int p(x)dx} \frac{du}{dx} - u e^{-\int p(x)dx} p(x) + p(x) u e^{-\int p(x)dx} = f(x)$$

$$\Rightarrow p e^{-\int p(x)dx} \frac{du}{dx} = f(x)$$

$$du = f(x) e^{\int p(x)dx} dx$$

$$u = \int f(x) e^{\int p(x)dx} dx + C$$

$$y = e^{-\int p(x)dx} \left[\int f(x) e^{\int p(x)dx} dx + C \right]$$

$$= \underbrace{e^{-\int p(x)dx} \int f(x) e^{\int p(x)dx} dx}_{\downarrow} + \underbrace{c e^{-\int p(x)dx}}_{\downarrow}$$

$C=0$

一个解

$$y(x) = y^*(x) + Y(x)$$

通解

$$y = e^{-\int_{x_0}^x p(\xi) d\xi} \left[\int_{x_0}^x f(\xi) e^{\int_{x_0}^{\xi} p(\zeta) d\zeta} d\xi + y_0 \right]$$

$$x \in (a, b)$$

$$x_0 \rightarrow x$$

$$\text{eg. } \frac{dy}{dx} - y \cot x = 2x \sin x$$

$$\frac{dy}{dx} - y \cot x = 0.$$

$$\ln|y| = \ln|\sin x| + \ln|c|$$

$$y = c \sin x$$

$$y = u \sin x$$

$$\frac{dy}{dx} = \sin x \frac{du}{dx} + u \cos x$$

$$\sin x \frac{du}{dx} = 2x \sin x.$$

$$\frac{du}{dx} = 2x$$

$$u = x^2 + c$$

$$y = (x^2 + c) \sin x \quad \leftarrow y|_{x=\frac{\pi}{2}} = 0.$$

$$c = -\frac{\pi^2}{4}$$

$$y = \left(x^2 - \frac{\pi^2}{4}\right) \sin x$$

$$2x \frac{dy}{dx} - y = -x$$

by

$$\frac{dy}{dx} - \frac{y}{2x} = -\frac{1}{2}$$

$$e^{\int \frac{1}{2x} dx} \left[-\frac{1}{2} \int e^{-\int \frac{1}{2x} dx} dx + C \right]$$

$$-\frac{1}{2} \ln|x|$$

$$\sqrt{|x|} \left[-\frac{1}{2} \int \frac{1}{\sqrt{|x|}} dx + C \right]$$

$$y = -x + C \sqrt{|x|}$$

eg. 方程 2π 周期解:

$$y(x) = y(x + 2\pi)$$

$$y(0) = y(2\pi)$$

= Bernoulli

$$\frac{dy}{dx} + p(x)y = f(x)y^n$$

$$y^{-n} \frac{dy}{dx} + p(x) y^{1-n} = f(x)$$

$$z = y^{1-n} \quad \rightarrow$$

$$\frac{dz}{dx} = (1-n) y^{-n} \frac{dy}{dx}$$

$$\frac{dz}{dx} + (1-n)p(x)z = (1-n)f(x)$$

eg. $\frac{dy}{dx} + \frac{y}{x} = a(\ln x) y^2$

$$y^{-2} \frac{dy}{dx} + \frac{1}{x} y^{-1} = a(\ln x)$$

$$z = y^{-1}$$

$$\frac{dz}{dx} = -y^{-2} \frac{dy}{dx}$$

$$= \frac{z}{x} - a \ln x \quad -\ln x$$

$$z = e^{-\int \frac{1}{x} dx} \left[\int -a \ln x e^{\int \frac{1}{x} dx} dx + C \right]$$

$$= x \left[\int (-a \ln x) \cdot \frac{1}{x} dx + C \right]$$

$$= x \left[C - \frac{a}{2} (\ln x)^2 \right]$$

$$y = \frac{1}{x \left[C - \frac{a}{2} (\ln x)^2 \right]}$$

eg. $\frac{dy}{dx} = \frac{y}{2x} + \frac{x^2}{2y}$

取 $n = -1$

$$y \frac{dy}{dx} = \frac{y^2}{2x} + \frac{x^2}{2}$$

$$z = y^2$$

$$\begin{aligned} \frac{dz}{dx} &= 2y \frac{dy}{dx} \\ &= \frac{y^2}{x} + x^2 \end{aligned}$$

$$\frac{dz}{dx} = \frac{z}{x} + x^2$$

$$\Rightarrow y^2 = z = cx + \frac{1}{2}x^3$$

三 Riccati $\frac{dy}{dx} = p(x)y^2 + q(x)y + r(x)$

取特解 $y = \phi_1(x)$

→ 一般不能用初等方法求解.

令 $u(x) = y(x) - \phi_1(x)$

$$\frac{du}{dx} = \frac{dy}{dx} - \frac{d\phi_1(x)}{dx}$$

$$= p(u + \phi_1)^2 + q(u + \phi_1) + r - p\phi_1^2 - q\phi_1 - r$$

$$= [2p\phi_1 + q]u(x) + p(x)u^2 \rightarrow \text{Bernoulli}$$

§4 全微分方程.

$$M(x, y)dx + N(x, y)dy = 0$$

$$M(x, y)dx + N(x, y)dy = du(x, y)$$

若 $y = y(x)$ 为一个解.

$$\begin{cases} M(x, y(x))dx + N(x, y(x))dy \equiv 0 \\ du(x, y(x)) \equiv 0 \\ \rightarrow u(x, y(x)) \equiv C_0 = u(x_0, y_0) \end{cases}$$

隐函数.

为全微分方程充要条件:

$$\frac{\partial M}{\partial y} \equiv \frac{\partial N}{\partial x}, (x, y) \in G.$$

法1

$$u(x, y) = \int_{(x_0, y_0)}^{(x, y)} M(x, y)dx + N(x, y)dy.$$

$$= \int_{x_0}^x M(\xi, y_0)d\xi + \int_{y_0}^y N(x, \eta)d\eta$$

法2

或 $du(x, y) = M(x, y)dx + N(x, y)dy$

$$\frac{\partial u}{\partial x} = M(x, y)$$

$$\frac{\partial v}{\partial y} = N(x, y)$$

$$u(x, y) = \int m(x, y) dx + p(y) \quad \checkmark C \quad \star$$

$$\frac{\partial v}{\partial y} = \frac{\partial}{\partial y} \int m(x, y) dx + p'(y) = N(x, y)$$

$$\Rightarrow p'(y) = N(x, y) - \frac{\partial}{\partial y} \int m(x, y) dx$$

法3 配凑

$$y dx + x dy = d(xy)$$

$$\frac{y dx - x dy}{y^2} = d\left(\frac{x}{y}\right)$$

$$\frac{-y dx + x dy}{x^2} = d\left(\frac{y}{x}\right)$$

$$\frac{-y dx + x dy}{x^2 + y^2} = d\left(\arctan \frac{y}{x}\right)$$

$$\frac{y dx - x dy}{x^2 - y^2} = d\left(\frac{1}{2} \ln \left| \frac{x-y}{x+y} \right| \right)$$

$$m(x, y) dx + N(x, y) dy = 0.$$

若 $\exists \mu(x, y) \neq 0$.

s.t. $\mu(x, y) m(x, y) dx + \mu(x, y) N(x, y) dy = 0$ 为全微分方程. ①

即 $\mu(x, y) \neq 0$.

$$\text{s.t. } \mu(x, y) M(x, y) dx + \mu(x, y) N(x, y) dy \equiv du(x, y)$$

则 $\mu(x, y)$ 为①的一个积分因子

$u(x, y) = C$ 为①通积分.

$$(y + x^2 + y^2) dx - x dy = 0.$$

$$\frac{y dx - x dy + \cancel{(x^2 + y^2) dx}}{x^2 + y^2} = 0$$

$$\text{对于 } M \frac{\partial u}{\partial y} + \mu \frac{\partial v}{\partial y} = N \frac{\partial v}{\partial x} + \mu \frac{\partial N}{\partial x}$$

若 $\mu(x, y)$ 仅与 x 或 y 有关

$$0 + \mu \frac{\partial v}{\partial y} = N \frac{\partial v}{\partial x} + \mu \frac{\partial N}{\partial x}$$

$$\mu \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = N \frac{d\mu}{dx}$$

$$\frac{1}{\mu} \cdot \frac{d\mu}{dx} = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}$$

$$\text{同理取 } \mu(y) \quad \frac{1}{\mu} \cdot \frac{d\mu}{dy} = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{-M}$$

$$\text{若 } \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} \text{ 仅与 } x \text{ 有关} = h(x)$$

$$\text{积分因子取 } \mu(x) = e^{\int h(x) dx}$$

$$\text{若 } \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{-M} \text{ 仅与 } y \text{ 有关} = h(y)$$

$$\text{积分因子取 } \mu(y) = e^{\int h(y) dy}$$

$$\text{eg. } \underbrace{(-y + xy^2)}_M dx + \underbrace{xy}_{N} dy = 0$$

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = -2(1 - xy)$$

$$\frac{-2 + 2xy}{-M} = \frac{2}{y} = \frac{2}{y}$$

$$\mu(y) = e^{\int \frac{2}{y} dy} = \frac{1}{y^2}$$

② 乘 $\mu(y)$

$$(-\frac{1}{y} + x) dx + \frac{x}{y^2} dy = 0.$$

$$\text{即 } x dx + \frac{xdy - ydx}{y^2} = 0.$$

$$\frac{x^2}{2} - \frac{x}{y} = C$$

§5. 高阶: 降阶

$$\therefore \frac{d^2 y}{dx^2} = f(x)$$

$$\frac{dy}{dx} = \int f(x) dx + C_1$$

$$y = \int (\int f(x) dx + C_1) dx + C_2$$

$$\therefore \frac{d^2 y}{dx^2} = f(x, \underbrace{\frac{dy}{dx}}_p)$$

$$\begin{cases} \frac{d^2 y}{dx^2} = f(x, y, \frac{dy}{dx}) \\ y|_{x=x_0} = y_0 \\ \frac{dy}{dx}|_{x=x_0} = y_0' \end{cases}$$

$$\frac{d^2y}{dx^2} = \frac{dp}{dx}$$

$$\frac{dp}{dx} = f(x, p) \Rightarrow p = \varphi(x, c_1)$$

$$\frac{dy}{dx} = \varphi(x, c_1)$$

$$\Rightarrow y = \int \varphi(x, c_1) dx + c_2$$

$$y^{(n)} = f(x, y^{(n-1)}) \Rightarrow y^{(n)} = \frac{dp}{dx}.$$

n次积分.

$$\equiv \frac{d^2y}{dx^2} = f(y, \frac{dy}{dx})$$

$$\frac{dy}{dx} \cdot \frac{dp}{dy}$$

$$\frac{dy}{dx} = p$$

$$\frac{dp}{dx} = f(y, p)$$

$$\frac{dp}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{dp}{dx} = \frac{dp}{dy}$$

$$\frac{dy}{dx} = p \frac{dp}{dy}$$

$$p \frac{dp}{dy} = f(y, p)$$

$$\text{eg. } \left(\frac{dy}{dx}\right)^2 - y \frac{d^2y}{dx^2} = 0$$

$$\text{取 } p = \frac{dy}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{dp}{dx} = p \frac{dp}{dy}$$

$$p^2 - yp \frac{dp}{dy} = 0.$$

$$y = c_2 e^{c_1 x}$$

U2

§1.

一般形式

$$\frac{d^n y}{dx^n} + p_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + p_n(x) y = f(x)$$

$f(x) \equiv 0$. 齐次线性微分方程

$$y^{(i)}|_{x=x_0} = y_0^{(i)}$$

$$y^{(i)}|_{x=x_0} = 0 \quad \text{唯一解 } y(x) \equiv 0.$$

$$L[y] \equiv \frac{d^n y}{dx^n} + p_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + p_n(x) y$$

$$2.1 \quad L[cy] = cL[y]$$

$$2.2 \quad L[y_1 + y_2] = L[y_1] + L[y_2]$$

$$2.3 \quad L\left[\sum_{i=1}^m c_i y_i\right] = \sum_{i=1}^m c_i L[y_i]$$

线性相关/无关,

$$a_1 y_1(x) + \dots + a_n y_n(x) \equiv 0 \quad \text{线性相关}$$

$$W(x) = \begin{vmatrix} y_1(x) & \cdots & y_m(x) \\ y_1'(x) & & y_m'(x) \\ \vdots & & \vdots \\ y_1^{(m-1)}(x) & & y_m^{(m-1)}(x) \end{vmatrix} = \phi(x)$$

为 $y_1(x) \dots y_m(x)$ 朗斯基行列式

齐次线性方程组不同解线性组合仍为解.

n 个线性无关解.

$y_1(x) \dots y_n(x)$ 为齐次方程 n 个解.

C_i 为常数. —————> 基本解组

$$y(x) = \sum_{i=1}^n C_i y_i(x) \text{ 为通解,}$$

$$y'' + p(x)y' + q(x)y = 0.$$

$$y(x) = C_1 y_1(x) + C_2 y_2(x)$$

非齐次.

① y_1^*, y_2^* 为非齐次解.

$\Rightarrow$ 则 $y_1^* - y_2^*$ 为齐次解.

② y^* 与 y 为非齐次解及对应解.

$y^* + y$ 也为解.

设 y^* : 非齐次解.

Y : 齐次方程通解.

$$y = Y + y^*.$$

$$L[y] = f_1(x) \quad L[y] = f_2(x)$$

$$L[y] = f_1(x) + f_2(x).$$

$$y = u(x) + i v(x)$$

$$L[y] = u(x) + i v(x)$$

§2 常系数线性微分方程.

一、二阶 $\quad$ 对于 $\frac{d^2 y}{dx^2} + p \frac{dy}{dx} + qy = 0.$

先求线性无关解 y_1, y_2

$$\text{s.t. } y = C_1 y_1 + C_2 y_2.$$

$$y = e^{\lambda x}$$

$$\lambda^2 e^{\lambda x} + p \lambda e^{\lambda x} + q e^{\lambda x} = 0.$$

$$\lambda^2 + p\lambda + q = 0 \quad \text{特征方程} \quad \lambda: \text{特征根.}$$

$$y_1 = e^{\lambda_1 x} \quad y_2 = e^{\lambda_2 x} \quad \text{两个解.}$$

① $\lambda_1 \neq \lambda_2.$

通解: $y = C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x}$

② $\lambda_1 = \lambda_2.$

$$p^2 - 4q = 0. \quad \lambda_1 = \lambda_2 = -\frac{p}{2}$$

$$y_1 = e^{\lambda_1 x}$$

$$\text{取 } y = u e^{\lambda_1 x}$$

$$\frac{dy}{dx} = (u' + \lambda_1 u) e^{\lambda_1 x}$$

$$\frac{d^2 y}{dx^2} = (u'' + 2\lambda_1 u' + \lambda_1^2 u) e^{\lambda_1 x}$$

$$u'' + (2\lambda_1 p)u' + (\lambda_1^2 + p\lambda_1 + q)u = 0.$$

$$\lambda_1 = \text{重根}$$

$$\Rightarrow \lambda_1^2 + p\lambda_1 + q = 0 \quad 2\lambda_1 p = 0.$$

$$u'' = 0.$$

$$\text{取 } u = x.$$

$$\text{通解: } y = (c_1 + c_2 x) e^{\lambda_1 x}$$

$$\textcircled{3} \quad \lambda_1 = \alpha + i\beta \quad \lambda_2 = \alpha - i\beta.$$

$$y_1 = e^{(\alpha + i\beta)x} \quad y_2 = e^{(\alpha - i\beta)x}$$

$$e^{(\alpha \pm i\beta)x} = e^{\alpha x} (\cos \beta x \pm i \sin \beta x)$$

$$\frac{1}{2} (y_1 + y_2) = e^{\alpha x} \cos \beta x$$

$$\frac{1}{2i} (y_1 - y_2) = e^{\alpha x} \sin \beta x$$

两个解.

$$\text{通解: } y = c_1 e^{\alpha x} \cos \beta x + c_2 e^{\alpha x} \sin \beta x \\ = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$$

$$\text{或} = e^{\alpha x} \sqrt{c_1^2 + c_2^2} \sin(\beta x + \varphi) \\ = A e^{\alpha x} \sin(\beta x + \varphi)$$

$$\text{其中 } A = \sqrt{c_1^2 + c_2^2} \\ \sin \varphi = \frac{c_2}{\sqrt{c_1^2 + c_2^2}} \\ \cos \varphi = \frac{c_1}{\sqrt{c_1^2 + c_2^2}}$$

特征方程 $\lambda^2 + p\lambda + q = 0$ 的根	微分方程 $\frac{d^2 y}{dx^2} + p \frac{dy}{dx} + qy = 0$ 的通解
有不相等的实根 $\lambda_1 \neq \lambda_2$	$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$
有相等的实根 $\lambda_1 = \lambda_2$	$y = c_1 e^{\lambda_1 x} + c_2 x e^{\lambda_2 x}$
有共轭复数根 $\lambda_1 = \alpha + i\beta$ $\lambda_2 = \alpha - i\beta$	$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$ 或 $y = A e^{\alpha x} \sin(\beta x + \varphi)$

$$\text{eg 1. } \frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} - 3y = 0.$$

$$\lambda^2 - 2\lambda - 3 = 0$$

$$\lambda_1 = 3 \quad \lambda_2 = -1$$

$$y = c_1 e^{3x} + c_2 e^{-x}$$

$$2. \frac{d^2 s}{dt^2} - 4 \frac{ds}{dt} + 4s = 0.$$

$$\lambda^2 - 4\lambda + 4 = 0 \quad \lambda_1 = \lambda_2 = 2$$

$$(C_1 + C_2 t) e^{2t}.$$

$$3. \frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 13y = 0.$$

$$\lambda^2 + 4\lambda + 13 = 0.$$

$$\lambda = -2 \pm 3i$$

$$y = e^{-2x} (C_1 \cos 3x + C_2 \sin 3x)$$

$$\text{或 } y = A e^{-2x} \sin(3x + \varphi)$$

二. n 阶.

$$L[y] \equiv y^{(n)} + p_1 y^{(n-1)} + \dots + p_n y = 0$$

$$D(\lambda) = \lambda^n + p_1 \lambda^{n-1} + \dots + p_n = 0.$$

$$\text{通解: } y = \sum_{i=1}^s \sum_{j=1}^{n_i} c_{ij} x^{j-1} e^{\lambda_i x}$$

其中特征根 $\lambda_1, \dots, \lambda_s$.

重数 $n_1, \dots, n_s, \sum n_i = n$

$$\lambda_s \downarrow \begin{matrix} e^{\lambda_1 x}, x e^{\lambda_1 x}, \dots, x^{n_1-1} e^{\lambda_1 x} \\ \dots \\ e^{\lambda_s x}, \dots, x^{n_s-1} e^{\lambda_s x} \end{matrix}$$

$$e^{\alpha x} \cos \beta x, x e^{\alpha x} \cos \beta x, \dots, x^{n_1-1} e^{\alpha x} \cos \beta x$$

$$e^{\alpha x} \sin \beta x, \dots, x^{n_1-1} e^{\alpha x} \sin \beta x$$

特征方程(2.21) 的根	微分方程(2.20) 通解中对应的项
① 单重实根 λ	对应一项 $ce^{\lambda x}$
② k 重实根 λ	对应 k 项 $(c_1 + c_2 x + \dots + c_k x^{k-1}) e^{\lambda x}$
③ 单重复数根 $\lambda_{1,2} = \alpha \pm \beta i, \beta > 0$	对应两项 $e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$
④ k 重复数根 $\lambda_{1,2} = \alpha \pm \beta i, \beta > 0$	对应 $2k$ 项 $e^{\alpha x} [(a_1 + a_2 x + \dots + a_k x^{k-1}) \cos \beta x + (b_1 + b_2 x + \dots + b_k x^{k-1}) \sin \beta x]$

eg1: $\frac{d^3 y}{dx^3} - 3 \frac{d^2 y}{dx^2} + 4y = 0$

$$\lambda^3 - 3\lambda^2 + 4 = (\lambda + 1)(\lambda - 2)^2 = 0.$$

$$y = c_1 e^{-x} + c_2 e^{2x} + c_3 x e^{2x}$$

eg2. $\frac{d^4 y}{dx^4} + k^4 y = 0.$

$$\lambda^4 + k^4 = 0. \Rightarrow (\lambda^2 + k^2 + \sqrt{2} \lambda k)(\lambda^2 + k^2 - \sqrt{2} \lambda k) = 0$$

$$+ \sqrt{2} k \lambda + k^2$$

$$\lambda = -\frac{\sqrt{2}}{2}k \pm i\frac{\sqrt{2}}{2}k \quad \lambda = \frac{\sqrt{2}}{2}k \pm i\frac{\sqrt{2}}{2}k$$

$$\text{三} \quad y'' + py' + qy = f(x)$$

$$\text{齐次} \quad y'' + py' + qy = 0$$

$$(-) f(x) = P_m(x) e^{\alpha x}$$

$$\frac{d^2 y}{dx^2} + p \frac{dy}{dx} + qy = P_m(x) e^{\alpha x}$$

$$\text{取 } y^* = Q(x) e^{\alpha x}$$

$\times q$

$Q(x)$ 多项式

$$\frac{dy^*}{dx} = \frac{dQ}{dx} e^{\alpha x} + \alpha Q e^{\alpha x} \quad \times p$$

$$\frac{d^2 y^*}{dx^2} = \frac{d^2 Q}{dx^2} e^{\alpha x} + 2\alpha \frac{dQ}{dx} e^{\alpha x} + \alpha^2 Q e^{\alpha x}$$

代回.

$$\lambda^2 + (2\alpha + p)\lambda + (\alpha^2 + p\alpha + q) = P_m(x)$$

$$\frac{d^2 Q}{dx^2} + (2\alpha + p) \frac{dQ}{dx} + (\alpha^2 + p\alpha + q) Q = P_m(x)$$

$\uparrow$
m次

$$\text{① } \alpha^2 + p\alpha + q \neq 0 \quad \alpha \text{ 不为根}$$

$$\text{令 } Q(x) = R_m(x) \quad \leftarrow m \text{ 次}$$

$$y^* = R_m(x) e^{\alpha x}$$

$$\text{② } \alpha^2 + p\alpha + q = 0 \quad 2\alpha + p \neq 0$$

α 单根.

$$\boxed{\lambda^2 + p\lambda + q = 0}$$

特征方程

$$\text{令 } Q(x) = x R_m(x)$$

$$y^* = x R_m(x) e^{\alpha x}$$

$$\Delta = p^2 - 4q$$

$$\omega^2$$

$$(3) \lambda^2 + p\lambda + q = 0, \quad 2\lambda + p = 0$$

$$\lambda = \text{重根}$$

$$\frac{d^2 Q}{dx^2} = p_m(x)$$

$$y^* = x^2 R_m(x) e^{\alpha x}$$

$$\text{总结为 } y^* = x^k R_m(x) e^{\alpha x}$$

$$k = \begin{cases} 0, & \lambda \text{ 不为特征根} \\ 1, & \lambda \text{ 单根} \\ 2, & \lambda \text{ 双根} \end{cases}$$

$$\text{eg. } \frac{d^2 y}{dx^2} + y = (x-2)e^{3x}$$

$$\lambda^2 + 1 \Rightarrow \lambda = \pm i$$

$$\frac{d^2 y}{dx^2} + y = 0 \quad : \quad y = C_1 \cos x + C_2 \sin x$$

$$\text{设 } y^* = (Ax+B)e^{3x}$$

$$(6+0)A + (3^2+1)(Ax+B) = x-2$$

$$2\lambda + p \quad \lambda^2 + p\lambda + q$$

$$A = \frac{1}{10} \quad B = -\frac{13}{50}$$

$$y^* = \left(\frac{1}{10}x - \frac{13}{50}\right)e^{3x}$$

$$y = Y + y^* = C_1 \cos x + C_2 \sin x + \left(\frac{1}{10}x - \frac{13}{50}\right)e^{3x}$$

eg. $y'' + py' + qy = 0$.

$$(C_1 + C_2 x)e^x$$

$$\lambda = 1$$

$$y'' - 2y' + y = x$$

$$y^* = Ax + B \quad \text{代入}.$$

$$A = 1 \quad B = 2$$

$$\therefore y = (C_1 + C_2 x)e^x + x + 2$$

$$(=) f(x) = P_m(x) e^{ax} \cos bx / \sin bx$$

$$P_m(x) e^{(a+ib)x} \text{ 实部}$$

$$\text{考虑 } y'' + py' + qy = P_n(x) e^{(a+ib)x}$$

$$\text{eg } y'' - y = e^x \sin 2x$$

$$y'' - y = 0 \quad \lambda = \pm 1$$

$$Y = C_1 e^x + C_2 e^{-x}$$

$\lambda = 1+2i$ 不为解

$$\text{设 } y = A e^{(1+2i)x}$$

$$A(1+2i)^2 - A = 1$$

$$A = -\frac{1}{8}(i+1)$$

$$y = -\frac{1}{8}(i+1)e^{(1+2i)x}$$

$$= -\frac{1}{8}(i+1)e^x (\cos 2x + i \sin 2x)$$

$$y^* = -\frac{1}{8}e^x (\cos 2x + i \sin 2x)$$

$$y = Y + y^* = C_1 e^x + C_2 e^{-x}$$

$$- \frac{1}{8}e^x (\cos 2x + i \sin 2x)$$

$$\text{又 } y'' + py' + qy = f(x)$$

$$y^* = x^k (R_n(x)e^{ax} \cos bx + S_n(x)e^{ax} \sin bx)$$

$$其中 h = \max\{m, L\}$$

$$k = \begin{cases} 0 & , a \pm bi \text{ 不为特征根} \\ 1 & , a \pm bi \text{ 为单特征根} \end{cases}$$

$$y'' + 4y = 2\cos 2x$$

$$\lambda^2 + 4$$

$$C_1 \cos 2x + C_2 \sin 2x \quad 0 \pm 2i$$

$$y^* = x(A \cos 2x + B \sin 2x)$$

$$\frac{d^2x}{dy^2} + (y + \sin^2 x) \left(\frac{dx}{dy} \right)^2.$$

$$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$$

$$\frac{d^2x}{dy^2} = - \frac{\frac{d^2y}{dx^2}}{\left(\frac{dy}{dx} \right)^3}$$

$$\frac{d^2y}{dx^2} - y = \sin^2 x = \frac{1}{2} - \frac{1}{2} \cos 2x$$

$$y = C_1 e^x + C_2 e^{-x}$$

$$\frac{1}{2} y_1^* = -\frac{1}{2}$$

$$-\frac{1}{2} \cos 2x; \quad y_2^* = -\frac{1}{2} + \frac{1}{10} \cos 2x$$

$$y = C_1 e^x + C_2 e^{-x} - \frac{1}{2} + \frac{1}{10} \cos 2x$$

推广

$$L[y] = f(x)$$

① 设 $f(x) = P_m(x) e^{\alpha x}$

$$k = \begin{cases} 3, & \alpha \text{ 为三重根} \\ \vdots \end{cases}$$

② 设 $f(x) = P_m(x) e^{\alpha x} \cosh bx + Q_t(x) e^{\alpha x} \sinh bx$

$$k = \begin{cases} 2, & a \pm bi = \text{重根} \\ \vdots \end{cases}$$

eg. $y''' + 3y'' + 3y' + y = (x-5)e^{-x}$

$$\lambda^3 + 3\lambda^2 + 3\lambda + 1 = 0$$

$$(\lambda+1)^3 = 0$$

$$\lambda = -1$$

$$\alpha = -1, \text{ 三重根}$$

$$Y = (C_1 + C_2 x + C_3 x^2) e^{-x}$$

$$\text{设 } y^* = x^3(Ax+B)e^{-x} \\ = (Ax^4+Bx^3)e^{-x}$$

$$\text{求 } y^{*'}, y^{*''}, y^{*'''} \quad \text{求}$$

$$\begin{cases} (24A-18B)x + 6B + 18Bx = x-5 \end{cases}$$

$$A = \frac{1}{24} \quad B = -\frac{5}{6}$$

$$y = Y + y^* = (C_1 + C_2x + C_3x^2 - \frac{5}{6}x^3 + \frac{1}{24}x^4)e^{-x}$$

$$\text{eg. } f(x) = e^{-x} + \frac{1}{2} \int_0^x (x-t)^2 f(t) dt$$

$$\frac{1}{2} \int_0^x (x^2 - 2xt + t^2) f(t) dt$$

$$= \frac{1}{2} x^2 \int_0^x f(t) dt - x \int_0^x t f(t) dt + \frac{1}{2} \int_0^x t^2 f(t) dt$$

$$f'(x) = -e^{-x} + x \int_0^x f(t) dt + \frac{x^2}{2} f(x) - x^2 f(x) - \int_0^x t f(t) dt + \frac{1}{2} x^2 f(x)$$

$$= -e^{-x} + x \int_0^x f(t) dt - \int_0^x t f(t) dt$$

$$f''(x) = e^{-x} + \int_0^x f(t) dt$$

$$f'''(x) = -e^{-x} + f(x)$$

$$f(0) = 1 \quad f'(0) = -1 \quad f''(0) = 1$$

$$f^{(4)}(x) - f(x) = -e^{-x}$$

$$\lambda^3 - 1 = 0$$

$$(\lambda - 1)(\lambda^2 + \lambda + 1) = 0$$

$$\lambda_1 = 1 \quad \lambda_{2,3} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$$

$$f(x) = c_1 e^x + e^{-\frac{1}{2}x} \left(c_2 \cos \frac{\sqrt{3}}{2}x + c_3 \sin \frac{\sqrt{3}}{2}x \right)$$

$$\text{取 } f^*(x) = Ae^{-x}$$

P 95

§4 欧拉方程

$$a_0 x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} x \frac{dy}{dx} + a_n y = f(x)$$

(-) 变量代换 $x = e^t \quad t = \ln x$

$$\text{eg. } a_0 x^2 \frac{d^2 y}{dx^2} + a_1 x \frac{dy}{dx} + a_2 y = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{1}{x} \cdot \frac{dy}{dt} \quad \begin{matrix} y = x^2 \\ t = \ln x \end{matrix}$$

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{1}{x} \cdot \frac{dy}{dt} \right)$$

$$= \frac{1}{x^2} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) \quad \frac{1}{x^2} (y'' - y')$$

$$\therefore a_0 \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) + a_1 \frac{dy}{dt} + a_2 y = f(e^t)$$

$$\text{即 } a_0 \frac{d^2 y}{dt^2} - (a_1 - a_0) \frac{dy}{dt} + a_2 y = f(e^t)$$

(二) 降阶

$$\frac{d^2 y}{dx^2} + p(x) \frac{dy}{dx} + q(x)y = 0.$$

若有非0解 y_1

引入 u . 取 $y = y_1 u$

$$y' = y_1 u' + y_1' u$$

$$y'' = y_1 u'' + 2y_1' u' + y_1 u''$$

$$\text{即 } y_1 u'' + [2y_1' + p(x)y_1]u' + \underbrace{[y_1'' + p(x)y_1' + q(x)y_1]}_{=0} u = 0.$$

$$\frac{1}{2} u' = z$$

$$\text{即 } y_1 \frac{dz}{dx} + [2y_1' + p(x)y_1]z = 0$$

$$z = \frac{C_2}{y_1^2} e^{-\int p(x) dx}$$

$$y = y_1 \left[C_1 + C_2 \int \frac{1}{y_1^2} e^{-\int p(x) dx} dx \right]$$

Liouville 公式

(三) 特殊

$$2p'(x) + p^2(x) - 4q(x) = 0$$

$$\frac{d^2 y}{dx^2} + p(x) \frac{dy}{dx} + q(x)y = 0.$$

$$y = uv \quad v = v(x)$$

$$u''v + u'(2v' + pv) + u(v'' + pv' + qu)$$

$$v: 2v' + pv = 0.$$

$$v = e^{-\int \frac{p}{2} dx}$$

$$u'' + pv' + qu = -\frac{1}{4}(2p' + p^2 - 4q)e^{-\int \frac{p}{2} dx}$$

$$= -\frac{a}{4}e^{-\int \frac{p}{2} dx}$$

$$u'' - \frac{a}{4}u = 0.$$

$$\text{eg. } 4 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + (x^2 + 1)y = 0.$$

$$\Rightarrow \frac{d^2 u}{dx^2} + \lambda u = 0.$$

$$\Rightarrow \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + \frac{1}{4}(x^2 + 1)y = 0.$$

$$p(x) = x \quad q(x) = \frac{1}{4}(x^2 + 1),$$

$$2p'(x) + p^2(x) - 4q(x) = 1$$

$$a = 1$$

$$\Rightarrow y = u e^{-\int \frac{p}{2} dx} = u e^{-\frac{x^2}{4}}$$

$$u'' - \frac{1}{4}u = 0$$

$$u = c_1 e^{\frac{x}{2}} + c_2 e^{-\frac{x}{2}}$$

$$y = e^{-\frac{x^2}{4}} (c_1 e^{\frac{x}{2}} + c_2 e^{-\frac{x}{2}})$$

二、变动任意常数

$$y'' + p_1(x)y' + p_2(x)y = f(x)$$

通解 $y = c_1 y_1(x) + c_2 y_2(x)$

$$y = u_1 y_1(x) + u_2 y_2(x)$$

①

$$\{ \cup u_1, u_2$$

$$y' = [u_1 y_1'(x) + u_2 y_2'(x)] + [u_1' y_1(x) + u_2' y_2(x)]$$

使 $u_1' y_1(x) + u_2' y_2(x) = 0$

$$R_1) y' = u_1 y_1'(x) + u_2 y_2'(x).$$

$$y'' = [u_1 y_1''(x) + u_2 y_2''(x)] + [u_1' y_1'(x) + u_2' y_2'(x)]$$

$$\begin{cases} u_1' y_1(x) + u_2' y_2(x) = 0 \\ u_1' y_1'(x) + u_2' y_2'(x) = f(x) \end{cases}$$

$$u_1' = - \frac{y_2(x)}{u_1(x)} f(x).$$

$$u_2' = \frac{y_1(x)}{w(x)} f(x).$$

$$u_1 = \int \frac{y_2(x)}{w(x)} f(x) dx$$

$\Rightarrow$

$$u_2 = \int \frac{y_1(x)}{w(x)} f(x) dx.$$

代入 ①

$$y^* = -y_1(x) \int \frac{y_2(x)}{w(x)} f(x) dx \\ + y_2(x) \int \frac{y_1(x)}{w(x)} f(x) dx$$

$$y = y_1(x) \left(C_1 - \int \frac{y_2(x)}{w(x)} f(x) dx \right)$$

$$+ y_2(x) \left(C_2 + \int \frac{y_1(x)}{w(x)} f(x) dx \right)$$

eg. 求 $y'' + a^2 y = f(x)$ 通解.

$$y = C_1 \cos ax + C_2 \sin ax$$

变力/生变/变力.

$$y = u_1 \cos ax + u_2 \sin ax$$

$$y' = -a u_1 \sin ax + a u_2 \cos ax \\ + u_1' \cos ax + u_2' \sin ax$$

$$\wedge \frac{1}{2} u_1' \cos ax + u_2' \sin ax = 0.$$

$$y' = -a u_1 \sin ax + a u_2 \cos ax$$

$$y'' = -a^2 u_1 \cos ax - a^2 u_2 \sin ax \\ - a u_1' \sin ax + a u_2' \cos ax$$

代入

$$- a u_1' \sin ax + a u_2' \cos ax = f(x)$$

$$a u_1' \cos ax + u_2' \sin ax = 0.$$

$$u_1' = -\frac{1}{a} f(x) \sin ax$$

$$u_2' = \frac{1}{a} f(x) \cos ax$$

$$u_1 = -\frac{1}{a} \int_{x_0}^x f(\xi) \sin a\xi d\xi$$

$$u_2 = \frac{1}{a} \int_{x_0}^x f(\xi) \cos a\xi d\xi$$

$$y^* = \frac{1}{a} \int_{x_0}^x f(\xi) \sin a(x-\xi) d\xi$$

$$y = C_1 \cos ax + C_2 \sin ax + \frac{1}{a} \int_{x_0}^x f(\xi) \sin a(x-\xi) d\xi$$

eg. $y'' + py' + qy = 0$ 通解 $(C_1 + C_2 x)e^x$

$$y_1(x) = e^x \quad y_2(x) = x e^x$$

$$w(x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} = \begin{vmatrix} e^x & x e^x \\ e^x & (1+x)e^x \end{vmatrix} = e^{2x}$$

$$y(x) = e^x (C_1 - \int x e^x \cdot x e^{-2x} dx)$$

$$+xe^x(C_2 + \int e^x \cdot xe^{-2x} dx)$$

$$=(C_1 + C_2 x)e^x + x + 2$$

$$y(0)=2 \quad y'(0)=0.$$

$$\Downarrow$$

特解. $y = -xe^x + x + 2.$

三. 幂级数.

$$y'' + y = 0$$

$$\text{设 } y = a_0 + a_1 x + \dots + a_n x^n + \dots$$

$$y' = a_1 + 2a_2 x + \dots + {}^{(n+1)}a_{n+1} x^n$$

$$y'' = 2 \cdot 1 a_2 + 3 \cdot 2 a_3 x + \dots + (n+2)(n+1) a_{n+2} x^n$$

$$(2a_2 + a_0) + (6a_3 + a_1)x + (12a_4 + a_2)x^2 + \dots$$

$$[(n+2)(n+1)a_{n+2} + a_n]x^n = 0$$

$$y = a_0 \left(1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 \dots \right)$$

$$+ a_1 \left(x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 \dots \right)$$

Maclaurin 级数.

$$a_0 \cos x + a_1 \sin x$$

$$y'' + p(x)y' + q(x)y = 0$$

$p(x), q(x)$ 在 $|x| < R$ 内:

$$p(x) = \sum_{n=0}^{\infty} p_n x^n \quad q(x) = \sum_{n=0}^{\infty} q_n x^n.$$

$$y = \sum_{n=0}^{\infty} a_n x^n.$$

但 ν 阶 Bessel 方程.

$$x^2 y'' + x y' + (x^2 - \nu^2) y = 0 \quad (\nu \geq 0). \text{ 不满足,}$$

定理.

$$\text{设 } x^2 y'' + p(x) y' + q(x) y = 0.$$

$$p(0) = 0.$$

$$y = x^c \sum_{k=0}^{\infty} a_k x^k$$

$$R = \infty$$

$$y = x^c \sum_{k=0}^{\infty} x^k = \sum_{k=0}^{\infty} a_k x^{k+c}$$

$$\Rightarrow (c^2 - \nu^2) a_0 x^c + [(c+1)^2 - \nu^2] a_1 x^{c+1} + \sum_{k=2}^{\infty} \{[(c+k)^2 - \nu^2] a_k + a_{k-2}\} x^{c+k} = 0$$

$$\begin{cases} a_0(c^2 - \nu^2) = 0 \\ a_1[(c+1)^2 - \nu^2] = 0 \\ [(c+k)^2 - \nu^2] a_k + a_{k-2} = 0. \end{cases}$$

$$\textcircled{1} c = \nu \Rightarrow a_1 = 0.$$

$$J_{\nu}(x) = \sum_{m=0}^{\infty} \frac{(-1)^m x^{\nu+2m}}{2^{2m} m! (\Gamma(\nu+1)) (\nu+1) \cdots (\nu+m)}$$

ν 阶第一类 Bessel 函数

$$\text{例 } J_\nu(x) = \sum_{m=0}^{\infty} (-1)^m \frac{x^{\nu+2m}}{2^{\nu+2m} m! \Gamma(\nu+m+1)}$$

$$\textcircled{2} C = -\nu.$$

$$J_{-\nu}(x) = \sum_{m=0}^{\infty} (-1)^m \frac{x^{-\nu+2m}}{2^{-\nu+2m} m! \Gamma(-\nu+m+1)}$$

$$y = C_1 J_\nu(x) + C_2 J_{-\nu}(x)$$

$\nu = 2 \Rightarrow J_2$ 求线性无关的另一个解

$$y_2(x) = \frac{J_2 \cos 2x - J_2(x)}{\sin 2x}$$

$$x^2 y'' + xy' + (x^2 - 2^2)y = 0.$$

第三章

$$\S 1 \quad \frac{dx_i}{dt} = f_i(t, x_1, \dots, x_n)$$

$$x_1(t) \dots x_n(t)$$

$$\frac{dx_i(t)}{dt} = f_i(t, x_1(t), \dots, x_n(t))$$

$$\frac{d^n x}{dt^n} = f(t, x, \frac{dx}{dt}, \dots, \frac{d^{n-1}x}{dt^{n-1}})$$

$$\text{令 } x = x_i \quad \frac{dx_i}{dt} = x_2 \quad \frac{dx_{n-1}}{dt} = x_n$$

$$\begin{cases} \frac{dx_1}{dt} = x_2 \\ \frac{dx_2}{dt} = x_3 \\ \vdots \\ \frac{dx_{n-1}}{dt} = x_n \end{cases}$$

$$\frac{dx_n}{dt} = f(t, x_1, \dots, x_n)$$

$$x_1(t_0) = x_{10}$$

$$x_n(t_0) = x_{n0}$$

$$x_i = \varphi_i(t, c_1, \dots, c_n)$$

$$\Phi_i(t, x_1, \dots, x_n; c_1, \dots, c_n) = 0$$

$$x_i = \varphi_i(t, c_1, \dots, c_n) \text{ 通积分}$$

$$\frac{dx_i}{dt} = \sum_{j=1}^n a_{ij}(t)x_j + f_i(t)$$

$$\frac{d^n x}{dt^n} + p_1(t) \frac{d^{n-1}x}{dt^{n-1}} + \dots + p_n(t)x = f(t) \quad (1)$$

$$\text{令 } x = x_1 \quad \frac{dx_1}{dt} = x_2 \quad \frac{dx_{n-1}}{dt} = x_n$$

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad \frac{dx}{dt} = \begin{bmatrix} \frac{dx_1}{dt} \\ \vdots \\ \frac{dx_n}{dt} \end{bmatrix} \quad f(t) = \begin{bmatrix} f_1(t) \\ \vdots \\ f_n(t) \end{bmatrix}$$

$$A(t) = \begin{bmatrix} a_{11}(t) & \dots & a_{1n}(t) \\ \vdots & & \vdots \\ a_{n1}(t) & \dots & a_{nn}(t) \end{bmatrix}$$

$$(1) \Rightarrow \frac{dx}{dt} = A(x) \times f(t)$$

$$f(t) \neq 0 \quad \text{非齐次}$$

$$\frac{dx}{dt} = A(t) \times \text{齐次}$$

$$x(t) = \begin{bmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{bmatrix}$$

§2.

$$\begin{cases} \frac{dx}{dt} = A(t)x + f(t) \\ x(t_0) = x_0 \end{cases}$$

$$\begin{cases} \frac{dx}{dt} = A(t)x \\ x(t_0) = 0. \end{cases}$$

$$x(t) \equiv 0.$$

0 解.

$$x = c_1 x_1(t) + \dots + c_m x_m(t)$$

$$\frac{d}{dt} \sum c_i x_i(t) = A(t) \sum c_i x_i(t)$$

$$= \sum c_i \frac{dx_i(t)}{dt} = \sum c_i A(t) x_i(t)$$

$$= \sum c_i \left[\frac{dx_i(t)}{dt} - A(t) x_i(t) \right] \\ \begin{matrix} 0 \\ 0 \end{matrix}$$

定义. $a_1 x_1(t) + \dots + a_m x_m(t) \equiv 0$. 线性相关.

若仅 $a_1 = \dots = a_m = 0$ 成立, 线性无关.

$\gamma_1 e^{\lambda_1 t} \dots \gamma_m e^{\lambda_m t}$. 线性无关.

$$a_1 \gamma_1 + a_2 \gamma_2 e^{(\lambda_2 - \lambda_1)t} + \dots + a_m \gamma_m e^{(\lambda_m - \lambda_1)t} \equiv 0$$

$$\lambda_i - \lambda_1 \neq 0.$$

$$\beta_m \gamma_m e^{\lambda_m t} \equiv 0.$$

$$x(t) = (x_1(t) \dots x_n(t))$$

$\det x(t)$. 朗斯基行列式

$$W(t) = \det X(t) = \begin{vmatrix} x_{11}(t) & \cdots & x_{1n}(t) \\ \vdots & & \vdots \\ x_{n1}(t) & \cdots & x_{nn}(t) \end{vmatrix}$$

$x_1(t) \cdots x_n(t)$. 齐次线性方程组 n 个解.

$x_1(t) \cdots x_n(t)$ 相关 $\Leftrightarrow W(t) \equiv 0$.

$$X(t) = \sum c_i x_i(t),$$

$x_i(t)$ 为线性无关解. 基本解组

基本矩阵

$$x(t) = X(t)C$$

$$W(t_0) = \begin{vmatrix} 1 & & 0 \\ & 1 & \\ 0 & & 1 \end{vmatrix} = 1 \neq 0.$$

$$x = x(t)C + x^*(t)$$

§3 常系数

$$\frac{dx}{dt} = Ax + f(t)$$

$$\frac{dx}{dt} = Ax$$

$$x = ve^{\lambda t}$$

$$\lambda e^{\lambda t} = A v e^{\lambda t}$$

$$E = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$$

$$(A - \lambda E)v = 0.$$

$$\det(A - \lambda E) = 0.$$

$$D(\lambda) = \begin{vmatrix} a_{11} - \lambda & \dots & a_{1n} \\ \vdots & a_{22} - \lambda & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} - \lambda \end{vmatrix} = 0.$$

$$x(t) = v_k e^{\lambda_k t}.$$

$\lambda = \lambda_k$, 特征根.

v_k 为 A 属于 λ_k 特征向量.

$$x(t) = v_k e^{\lambda_k t}$$

(-) 单根

$$v_1 e^{\lambda_1 t}, \dots, v_n e^{\lambda_n t}$$

$$x = \sum c_i v_i e^{\lambda_i t}$$

eg.
$$\begin{cases} \frac{dx_1}{dt} = -3x_1 + 4x_2 - 2x_3, \\ \frac{dx_2}{dt} = x_1 + x_3, \\ \frac{dx_3}{dt} = 6x_1 - 6x_2 + 5x_3. \end{cases}$$

$$\begin{vmatrix} -3-\lambda & 4 & -2 \\ 1 & -\lambda & 1 \\ 6 & -6 & 5-\lambda \end{vmatrix}$$

$$= -(\lambda-2)(\lambda-1)(\lambda+1) = 0.$$

$\lambda_1=2/\lambda_2=1/\lambda_3=-1$ 单根.

$$\text{设 } \lambda_1=2 \quad v_1 = \begin{bmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \end{bmatrix}$$

$$\begin{bmatrix} -3-\lambda_1 & 4 & -2 \\ 1 & -\lambda_1 & 1 \\ 6 & -6 & 5-\lambda_1 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \end{bmatrix} = 0.$$

$$\begin{bmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} e^{2t}$$

$$\lambda_2: \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \lambda_3: \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = c_1 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^t + c_3 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} e^{-t}$$

若为复数根

$$\lambda_1 = \alpha + i\beta \quad \lambda_2 = \alpha - i\beta$$

$$v_1 = p + iq \quad v_2 = \bar{v}_1 = p - iq$$

$$v_1(t) = (p + iq)e^{(\alpha + i\beta)t}$$

$$= e^{\alpha t} (p \cos \beta t - q \sin \beta t) + i e^{\alpha t} (p \sin \beta t + q \cos \beta t)$$

$$x_2(t) = (p - iq) e^{(\alpha - i\beta)t}$$

$$= e^{\alpha t} (p \cos \beta t - q \sin \beta t) - i e^{\alpha t} (p \sin \beta t + q \cos \beta t)$$

↙ ↘
两个实值解.

eg. $\frac{d}{dt} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & -1 \\ -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$

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$$D(\lambda) = \begin{vmatrix} 2-\lambda & 1 & 0 \\ 1 & 3-\lambda & -1 \\ -1 & 2 & 3-\lambda \end{vmatrix}$$

$$= -(\lambda-2)(\lambda^2-6\lambda+10)$$

$$\lambda_1 = 2 \quad \lambda_{2,3} = 3 \pm i$$

$$\begin{vmatrix} \cos & -\sin \\ \sin & \cos \end{vmatrix}$$

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\alpha = 3 \quad \beta = 1$$

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{2t}$$

$$\begin{bmatrix} 1 \\ 1+i \\ 2-i \end{bmatrix} e^{(3+i)t}$$

$$= \begin{bmatrix} \cos t \\ \cos t - \sin t \\ 2 \cos t + \sin t \end{bmatrix} e^{3t} + i \begin{bmatrix} \sin t \\ \cos t + \sin t \\ 2 \sin t - \cos t \end{bmatrix} e^{3t}$$

$$C_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{2t} + C_2 \begin{bmatrix} \cos t \\ \cos t - \sin t \\ 2\cos t + \sin t \end{bmatrix} e^{3t} + C_3 \begin{bmatrix} \sin t \\ \cos t + \sin t \\ 2\sin t - \cos t \end{bmatrix} e^{3t}$$

(二) 有重根

$$x(t) = (v_0 + \frac{t}{1!} v_1 + \frac{t^2}{2!} v_2 + \dots + \frac{t^{k-1}}{(k-1)!} v_{k-1}) e^{\lambda_0 t}$$

$$(A - \lambda_0 E)^k v_0 = 0$$

$$\frac{dx}{dt} = Ax$$

$$\lambda_1, \dots, \lambda_s \quad n_1 + \dots + n_s = n$$

$$p_i(t) e^{\lambda_i t}$$

$p_i(t)$ 向量函数.

$$x_i = \sum_{j=1}^s \sum_{l=1}^{n_j} c_{ij} p_j(t) e^{\lambda_j t}$$

例 3 解方程组

eg. $\frac{dx}{dt} = Ax, x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 4 \\ 1 & 0 & -4 \end{bmatrix}.$

$$\det(A - \lambda E)$$

$$= \begin{vmatrix} -1-\lambda & 1 & 0 \\ 0 & -1-\lambda & 4 \\ 1 & 0 & -4-\lambda \end{vmatrix}$$

$$= -\lambda(\lambda+3)^2 = 0$$

$$\lambda_1 = 0 \quad \lambda_2 = -3 \text{ 重根}$$

$$\lambda_1 = 0$$

$$V_0 = \begin{bmatrix} 4 \\ 4 \\ 1 \end{bmatrix}$$

$$(A+3E)^2 V_0 = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 4 \\ 1 & 0 & -1 \end{bmatrix}^2 \begin{bmatrix} 4 \\ 4 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 4 & 4 \\ 4 & 4 & 4 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 4 \\ 1 \end{bmatrix} = 0$$

$$V_0^{(1)} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad V_0^{(2)} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

$$V_1^{(1)} = (A+3E) V_0^{(1)} \\ = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 4 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_2 = \left[\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + t \begin{bmatrix} -2 \\ -4 \\ 2 \end{bmatrix} \right] e^{-3t} (V_0^{(1)} + t V_1^{(1)}) e^{-3t}$$

$$V_1^{(2)} = (A+3E) V_0^{(2)} \\ = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 4 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_3 = \left[\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + t \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix} \right] e^{-3t}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = c_1 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_1 + c_2 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_2 + c_3 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_3 \\ = c_1 \begin{bmatrix} 4 \\ 4 \\ 1 \end{bmatrix} + c_2 \left[\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + t \begin{bmatrix} -2 \\ -4 \\ 2 \end{bmatrix} \right] e^{-3t} \\ + c_3 \left[\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + t \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix} \right] e^{-3t}$$

eg.
$$\begin{cases} \frac{dx}{dt} = -x + y + z \\ \frac{dy}{dt} = x - y + z, \\ \frac{dz}{dt} = x + y - z. \end{cases}$$

$$\begin{vmatrix} -1-\lambda & 1 & 1 \\ 1 & -1-\lambda & 1 \\ 1 & 1 & -1-\lambda \end{vmatrix}$$

$$-(\lambda-1)(\lambda+2)^2$$

$$\begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} e^t \quad \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

二. 非齐次

变通任意常数法

$$\frac{dx}{dt} = A(t)x + f(t)$$

$$\frac{dx}{dt} = A(t)x$$

$$x = \chi(t) c.$$

$$x = \chi(t) c(t).$$

$$\frac{d(x)}{dt} = \frac{d\chi(t)}{dt} c(t) + \chi(t) \frac{d(c(t))}{dt}.$$

$$= A(t) \chi(t) c + \chi(t) \frac{d(c(t))}{dt}.$$

$$\chi(t) c'(t) = f(t)$$

$$c'(t) = X^{-1}(t) = f(t)$$

$$c(t) = \int_{t_0}^t X^{-1}(\tau) f(\tau) d\tau$$

$$c(t_0) = 0$$

$$x^* = X(t) = \int_{t_0}^t X^{-1}(\tau) f(\tau) d\tau$$

$$x(t) = X(t)\vec{c} + X(t) \int_{t_0}^t X^{-1}(\tau) f(\tau) d\tau$$

其中 $\vec{c}$ 为 n 维任意常向量.

$$t_0 \in (a, b)$$

$$\therefore \text{pre. } x(t_0) = x_0. \text{ 代入}$$

$$\Rightarrow \vec{c} = X^{-1}(t_0)x_0.$$

$$\begin{cases} \frac{dx}{dt} = A(t)x + f(t), \\ x(t_0) = x_0, t_0 \in (a, b) \end{cases}$$

$$\text{sol. } x(t) = X(t)X^{-1}(t_0)x_0 + X(t) \int_{t_0}^t X^{-1}(\tau) f(\tau) d\tau$$

$$t_0 \in (a, b)$$

例 6 试解初值问题

eg. $\frac{dx}{dt} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & -2 \\ 3 & 2 & 1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ e^t \cos 2t \end{bmatrix},$

$$x(0) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

$$\begin{vmatrix} 1-\lambda & 0 & 0 \\ 2 & 1-\lambda & -2 \\ 3 & 2 & 1-\lambda \end{vmatrix} = (1-\lambda)(\lambda^2 + 2\lambda + 5) = 0$$

$$\lambda_1 = 1, \lambda_2 = 1+2i, \lambda_3 = 1-2i.$$

$$\Downarrow$$

$$\begin{bmatrix} 2 \\ -3 \\ 2 \end{bmatrix} \quad x_1 = \begin{bmatrix} 2 \\ -3 \\ 2 \end{bmatrix} e^t$$

$$\lambda_2 = v_2 = \begin{bmatrix} 0 \\ 1 \\ -i \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 0 \\ 1 \\ -i \end{bmatrix} e^{(1+2i)t}.$$

$$= e^t \begin{bmatrix} 0 \\ 1 \\ -i \end{bmatrix} (\cos 2t + i \sin 2t)$$

$$= e^t \left[\begin{bmatrix} 0 \\ \cos 2t \\ \sin 2t \end{bmatrix} + i \begin{bmatrix} 0 \\ \sin 2t \\ -\cos 2t \end{bmatrix} \right]$$

$$x(t) = e^t \begin{bmatrix} 2 & 0 & 0 \\ -3 & \cos 2t & \sin 2t \\ 2 & \sin 2t & -\cos 2t \end{bmatrix}$$

$$e^t \begin{bmatrix} 2 & 0 & 0 \\ -3 & \cos 2t & \sin 2t \\ 2 & \sin 2t & -\cos 2t \end{bmatrix} c'(t) = \begin{bmatrix} 0 \\ 0 \\ e^t \cos 2t \end{bmatrix}$$

$$C'(t) = \begin{bmatrix} 0 \\ \sin 2t \cos 2t \\ -\cos^2 2t \end{bmatrix}$$

$$C = C' \int_0^t \begin{bmatrix} 0 \\ \sin 2\tau \cos 2\tau \\ -\cos^2 2\tau \end{bmatrix} d\tau$$

$$= Ct \begin{bmatrix} 0 \\ \frac{1}{8}(1 - \cos 4t) \\ -\frac{1}{2} - \frac{1}{8}\sin 4t \end{bmatrix}$$

$$\Rightarrow x(t) = e^t \begin{bmatrix} 2 & 0 & 0 \\ -3 & \cos 2t & \sin 2t \\ 2 & \sin 2t & -\cos 2t \end{bmatrix} \left[c + \begin{bmatrix} 0 \\ \frac{1}{8}(1 - \cos 4t) \\ -\frac{t}{2} - \frac{1}{8}\sin 4t \end{bmatrix} \right].$$

再由 $x(0) = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, 得

$$\begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 2 & 0 & -1 \end{bmatrix} c = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

所以

$$c = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}.$$

于是所给初值问题的解是

$$\begin{aligned} x(t) &= e^t \begin{bmatrix} 2 & 0 & 0 \\ -3 & \cos 2t & \sin 2t \\ 2 & \sin 2t & -\cos 2t \end{bmatrix} \begin{bmatrix} 0 \\ \frac{1}{8}(1 - \cos 4t) \\ -1 - \frac{t}{2} - \frac{1}{8}\sin 4t \end{bmatrix} \\ &= e^t \begin{bmatrix} 0 \\ \cos 2t - \left(1 + \frac{t}{2}\right) \sin 2t \\ \left(1 + \frac{t}{2}\right) \cos 2t + \frac{5}{4} \sin 2t \end{bmatrix}. \end{aligned}$$

eg.

例 7 求方程组

$$\begin{cases} \frac{dx}{dt} = x + 2y + e^t, \\ \frac{dy}{dt} = 4x + 3y \end{cases} \rightarrow y = \frac{1}{2} \left(\frac{dx}{dt} - x - e^t \right)$$

$$\frac{1}{2} \frac{d}{dt} \left(\frac{dx}{dt} - x - e^t \right) = 4x + \frac{3}{2} \left(\frac{dx}{dt} - x - e^t \right)$$

$$\frac{d^2x}{dt^2} - 4 \frac{dx}{dt} - 5x = -2e^t$$

$$x = c_1 e^{5t} + c_2 e^{-t} + \frac{1}{4} e^t$$

$$\Rightarrow y = 2c_1 e^{5t} - c_2 e^{-t} - \frac{1}{2} e^t$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{5t} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t} + \begin{pmatrix} \frac{1}{4} \\ -\frac{1}{2} \end{pmatrix} e^t.$$

例 8 求方程组

eg.
$$\begin{cases} 2 \frac{dx}{dt} + \frac{dy}{dt} + y - t = 0, \\ \frac{dx}{dt} + \frac{dy}{dt} - x - y - 2t = 0 \end{cases}$$

$$\frac{dx}{dt} + x + 2y + t = 0.$$

$$\Rightarrow y = -\frac{1}{2} \left(\frac{dx}{dt} + x + t \right)$$

$$\text{代入} \Rightarrow \frac{d^2x}{dt^2} - 2 \frac{dx}{dt} + x = -3t - 1,$$

